Exact Prime Covers and Arithmetic Obstructions

Salem powers, repeated blocks, Euler integrals and Lambert functions

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Abstract

We assemble a source-audited collection of exact classifications and arithmetic obstructions, retaining complete proofs and the original target constants. For the largest root τ of $X^4 - X^3 - X^2 - X + 1$, every $\lfloor \tau^{hn} \rfloor$ has a proper divisor in $\{2, 3, 5\}$ if and only if $260 \mid h$; the natural density of terms coprime to 30 is calculated for every exponent. Further results exclude macroscopic repetitions in three arithmetic words, classify logarithmic contamination for rational floor kernels, and show why the raw twin-prime Euler products have approximation exponent zero relative to their reduced denominators. An asymmetric Euler integral admits exact integer linear forms and short cross-index logarithm elimination; equally short identically zero forms demonstrate the unresolved nonvanishing requirement. Finally, a fixed-base Lambert family representing the Erdős–Borwein and reciprocal Fibonacci constants has an exact all-jet orbit formula but admits no finite rational power-Mahler system. The manuscript consolidates selected HTML results with their corpus antecedents and the preceding Salem classification. It claims neither historical priority nor a new transcendence theorem for an unresolved named constant.

Contents

1 Purpose, selection and mathematical scope **2**

2 The exact prime-cover classification **2**

3 The conjugate correction and its distribution **3**

4 A finite criterion for a fixed exponent **4**

5 The quartic: exact arithmetic and proof of the main theorem **5**

6 The exact density classification **6**

7 Macroscopic repetitions in three arithmetic words **6**

8 A classification of logarithmic contamination **8**

9 The raw twin-prime Euler-product approximation **10**

10 Cross-index elimination for an asymmetric Euler integral **11**

11 Exact coordinates and finite-system obstructions for two reciprocal sums **14**

12 Consequences and remaining research tasks **18**

A A self-contained exact arithmetic certificate **19**

B Source concordance for the selected results **21**

1 Purpose, selection and mathematical scope

The starting materials are the supplied HTML research record [1], the preceding fixed-exponent Salem note [2], and the 49 TeX manuscripts in the repository snapshot [3]. The selection is mathematical rather than chronological: we retain results that give an exact classification, preserve an original constant through a useful coordinate system, or isolate a demonstrable failure of a proposed arithmetic argument. All selected statements are proved here from their stated standard inputs.

Four different conclusions must be distinguished. A complete finite classification can decide every exponent of a specified prime-cover problem. A functional obstruction can exclude a specified system of functional equations. An exact elimination identity can produce small integer linear forms. None of the latter conclusions, by itself, establishes transcendence of the value that motivated it. In particular, a nonzero coefficient vector can produce the zero linear form.

The prime-cover classification in Theorem 2.1 and its general criterion in Theorem 4.1 constitute the main positive classification. The other sections provide complementary information about repeated arithmetic data, logarithmic contamination, denominator growth and finite-dimensional functional methods. These are distinct mechanisms; their integration is not a claim that one new principle solves all of the underlying constant problems.

The comparison used the immutable commit

`f70e5fedabf7b90313f703eb85f7dff72a055541`.

A result described as additional to that snapshot is not thereby historically new. The source map in Appendix B identifies the retained HTML items, overlapping corpus statements and the specific refinements included here. The repository describes its contents as research drafts; source inclusion is not treated as independent certification of a theorem. Likewise, the exact arithmetic certificate in Appendix A verifies the finite computations, while the analytic arguments in the text justify their infinite-index consequences.

2 The exact prime-cover classification

A finite set of primes S *covers* a positive integer sequence if every term has a proper prime divisor in S. An *eventual cover* allows finitely many initial exceptions. Covering is stronger than compositeness: a composite sequence need not admit any fixed finite cover.

The supplied manuscripts [6, 9] establish effective powers of Salem numbers with covered floors and classify the prime sets that work after choosing a suitable power. Here the exponent is prescribed, so the trace need not have its identity residue. The following specific classification is the principal positive classification of this paper.

Theorem 2.1. *Let $\tau > 1$ be the largest root of*

$$P(X) = X^4 - X^3 - X^2 - X + 1.$$

For every integer $h \geq 1$, the following conditions are equivalent:

1. *For every $n \geq 1$, the integer $\left\lfloor \tau^{hn} \right\rfloor$ has a proper prime divisor in $\{2, 3, 5\}$.*
2. *The same holds for all sufficiently large n.*
3. *$260 \mid h$.*

The natural density

$$\delta(h) = \lim_{N\to\infty} \frac{1}{N} \#\{1 \leq n \leq N : \gcd(30, \left\lfloor \tau^{hn} \right\rfloor) = 1\}$$

exists and is given by Table 2. In particular,

$$260 \nmid h \quad \Longrightarrow \quad \delta(h) \geq \frac{1}{12},$$

with equality precisely when $\gcd(h, 260) = 130$.

This makes 260 the least exponent for the indicated three-prime cover. It does not make 260 the least exponent for compositeness by an arbitrary mechanism, or for coverage by a larger finite prime set. In particular, positive density of terms coprime to 30 is not a statement about prime terms.

The surrounding literature includes constructions of composite power floors and the explicit-example questions of Hahn–Ismailescu–Kim–Kim [10]. Those questions motivate the setting; this paper does not present their transcendental-number question as newly solved.

3 The conjugate correction and its distribution

We use the convention that a Salem number has degree at least four. Let τ have degree $d = 2s + 2$, with conjugates

$$\tau, \tau^{-1}, z_1, z_1^{-1}, \ldots, z_s, z_s^{-1}, \qquad |z_j| = 1.$$

Choose the z_j one from each reciprocal pair. Define

$$T_m = \mathrm{Tr}_{\mathbb{Q}(\tau)/\mathbb{Q}}(\tau^m) \in \mathbb{Z}, \qquad E_m = \tau^{-m} + \sum_{j=1}^{s} (z_j^m + z_j^{-m}).$$

The exact identity, including integral correction values, is

$$\lfloor \tau^m \rfloor = T_m - \lceil E_m \rceil . \tag{1}$$

We reproduce the needed analytic facts so that the finite calculation is not asked to stand in for the infinite argument.

Lemma 3.1 (Strict correction window)**.** *For every* $m \geq 1$,

$$-(d-2) < E_m < d - 2, \qquad \lceil E_m \rceil \in Q_d := \{-d+3, \ldots, d-2\}.$$

Proof. None of the z_j is a root of unity: otherwise the irreducible minimal polynomial of τ would be cyclotomic. Put $u = \tau^{-m} \in (0, 1)$ and $x_j = |1 - z_j^m|^2 > 0$. The nonzero algebraic-integer norm satisfies

$$1 \leq \left| N_{\mathbb{Q}(\tau)/\mathbb{Q}}(\tau^m - 1) \right| = \frac{(1-u)^2}{u} \prod_{j=1}^{s} x_j.$$

Thus $\prod_j x_j \geq u/(1-u)^2 > u$. If $\sum_j x_j \leq u$, the arithmetic–geometric mean inequality would give

$$\prod_j x_j \leq (u/s)^s \leq u,$$

a contradiction. Since $E_m = u + 2s - \sum_j x_j$, the upper bound follows. The lower bound follows from $E_m \geq u - 2s > -2s$. Taking ceilings gives the stated finite integer set. □

Lemma 3.2 (All corrections on every progression)**.** *For each* $q \in Q_d$*, there is a real number* $w_q > 0$ *such that, on every fixed positive-step arithmetic progression of* m*, the condition* $\lceil E_m \rceil = q$ *has relative natural density* w_q*. The weights are independent of the progression. In degree four,*

$$(w_{-1}, w_0, w_1, w_2) = \left(\frac{1}{3}, \frac{1}{6}, \frac{1}{6}, \frac{1}{3} \right). \tag{2}$$

Proof. Write $z_j = e^{2\pi i\theta_j}$. First, $1, \theta_1, \dots, \theta_s$ are linearly independent over $\mathbb{Q}$. More generally suppose $\prod_j z_j^{a_j} = \omega$, where the a_j are integers and ω is a root of unity. For a fixed j, choose an automorphism of the splitting field sending z_j to τ. No other selected z_k can map to τ by injectivity, or to τ^{-1} because that would force $z_k = z_j^{-1}$. Every other image therefore has modulus one. Taking absolute values after applying the automorphism yields $\tau^{a_j} = 1$, so $a_j = 0$. Repeating this proves the assertion.

For $m = an + b$, $a \geq 1$, every nontrivial integral character of the vector $(m\theta_1, \dots, m\theta_s)$ has irrational frequency. Its normalized exponential sum is a bounded geometric sum divided by its length and tends to zero. The Weyl criterion therefore gives equidistribution on the s-torus.

On that torus let

$$Y(t) = 2\sum_{j=1}^{s} \cos(2\pi t_j), \qquad w_q = \operatorname{meas}\{t : q-1 < Y(t) < q\}.$$

The interval $(q-1, q)$ meets the interior of the range $[-2s, 2s]$, and its inverse image contains a nonempty open set. Hence $w_q > 0$. Each level set $Y = c$ has measure zero: after fixing $t_2, \dots, t_s$, there are at most two possible values of t_1, so Fubini's theorem applies. Equidistribution now gives the frequencies. The vanishing term τ^{-m} does not change them, since neighborhoods of the two boundary level sets have measures tending to zero. In degree four the four arc lengths for $2\cos(2\pi t)$ give (2). □

4 A finite criterion for a fixed exponent

Let C be an integral companion matrix of the minimal polynomial of τ. It is invertible modulo every positive integer, since a Salem number is a unit, and $T_m = \operatorname{tr}(C^m)$.

Theorem 4.1 (Fixed-exponent criterion)**.** *Let S be a finite nonempty prime set, $M = \prod_{p\in S} p$, and $J \subset \mathbb{Z}$ finite and nonempty. Fix $h \geq 1$ and put*

$$L = \operatorname{ord}(C^h \bmod M), \qquad s_k = \operatorname{tr}(C^{hk}) \bmod M \quad (0 \leq k < L).$$

For all sufficiently large n, every $\left\lfloor \tau^{hn} \right\rfloor + j$, $j \in J$, has a proper prime divisor in S if and only if

$$\gcd(M, s_k - q + j) > 1 \quad (0 \leq k < L,\ q \in Q_d,\ j \in J). \tag{3}$$

For each fixed j, the natural density of indices with $\gcd(M, \left\lfloor \tau^{hn} \right\rfloor + j) = 1$ is

$$D_{h,j} = \frac{1}{L}\sum_{k=0}^{L-1} \sum_{q\in Q_d} w_q \mathbf{1}_{\gcd(M, s_k - q + j) = 1}. \tag{4}$$

Proof. If (3) holds, (1) and Lemma 3.1 give a prime divisor in S for every term. For sufficiently large n, every term is larger than $\max S$, so the divisor is proper.

If the condition fails at (k, q, j), restrict to positive indices $n \equiv k \pmod L$. The trace is s_k modulo M. Lemma 3.2 supplies a positive relative-density subset on which $\lceil E_{hn} \rceil = q$. On that subset (1) makes the shifted floor coprime to M, contradicting eventual coverage. Finally, split all indices into their L residue progressions and apply the same frequency statement to each; the resulting finite sum is (4). □

Corollary 4.2 (Finite search over exponents)**.** *Put $o = \operatorname{ord}(C \bmod M)$. The criterion and densities of Theorem 4.1 depend on h only through $g = \gcd(h, o)$. If any exponent gives an eventual cover, the least such exponent is the smallest divisor g of o whose trace orbit passes* (3)*.*

Proof. The cyclic subgroups generated by C^h and C^g are equal. Their elements, and hence their traces with multiplicity, differ only by a permutation. Moreover $g \le h$ and g itself is an admissible exponent whenever h is. □

A smaller scalar period of $(T_m \bmod M)$ may replace the matrix order in a particular example. The quartic below has scalar period 260, although its companion matrix has order 780 modulo 30.

5 The quartic: exact arithmetic and proof of the main theorem

The polynomial P in Theorem 2.1 is irreducible modulo two. It has no root in $\mathbb{F}_2$, and it is not $(X^2+X+1)^2$, the only possible product of irreducible quadratic polynomials in degree four over that field. Thus P is irreducible over $\mathbb{Q}$.

Dividing by X^2 and putting $y = X + X^{-1}$ gives $y^2 - y - 3$. Its roots are $(1 \pm \sqrt{13})/2$, one above 2 and one in $(-2, 2)$. Consequently τ is a quartic Salem number. Comparing $\tau + \tau^{-1} = (1+\sqrt{13})/2$ with $3/2 + 2/3$ also gives $\tau > 3/2$.

The traces satisfy

$$\begin{aligned} &T_0 = 4, \quad T_1 = 1, \quad T_2 = 3, \quad T_3 = 7, \\ &T_{m+4} = T_{m+3} + T_{m+2} + T_{m+1} - T_m. \end{aligned} \tag{5}$$

The integer calculations used below are reproduced by the standard-library Python certificate embedded in Appendix A. They can also be obtained directly from (5). No floating-point approximation is used.

The four-term state modulo 30 returns to its initial state at 260. It does not return at $260/p$ for $p = 2, 5, 13$. The recurrence transition is invertible modulo 30, because its trailing coefficient is -1. These tests prove that the scalar period is exactly 260: the period divides 260, and any proper divisor would divide $260/p$ for at least one prime divisor p of 260. An independent matrix calculation verifies order 780 for

$$C = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} \pmod{30}.$$

Specifically, $C^{780} = I$ and $C^{780/p} \neq I$ for $p = 2, 3, 5, 13$.

For $Q_4 = \{-1, 0, 1, 2\}$, define a trace residue t to be safe when

$$\gcd(30, t - q) > 1 \quad \text{for every } q \in Q_4.$$

The safe residues modulo 30 are exactly $4, 5, 26, 27$. Every proper divisor g of 260 has an unsafe trace T_g, as shown in Table 1.

g	1	2	4	5	10	13	20	26	52	65	130
$T_g \bmod 30$	1	3	7	16	18	1	22	9	19	16	24

Table 1: All proper divisors of the trace period yield unsafe traces.

Proof of Theorem 2.1, equivalences. If $260 \nmid h$, put $g = \gcd(h, 260) < 260$. There is an integer n_0 with $hn_0 \equiv g \pmod{260}$, because h/g is invertible modulo $260/g$. Along the corresponding progression of positive n, the trace is T_g modulo 30. Table 1 supplies an unsafe trace; Lemma 3.2 supplies a positive relative density of a correction q making the floor coprime to 30. Eventual coverage therefore fails.

If $260 \mid h$, every trace T_{hn} equals $T_0 = 4$ modulo 30. The correction window gives floor residues $2, 3, 4, 5$, all divisible by one of $2, 3, 5$. Since $h \geq 260$ and $\tau > 3/2$, every floor is greater than 5, making the divisor proper for every $n \geq 1$. This proves all three equivalences. □

6 The exact density classification

Let $g = \gcd(h, 260)$ and $\ell = 260/g$. Multiplication by h/g permutes the residue classes modulo ℓ. Combining the scalar trace period with (2) and (4) gives

$$\delta(h) = \frac{1}{\ell} \sum_{a=0}^{\ell-1} \sum_{q=-1}^{2} w_q \mathbf{1}_{\gcd(30, T_{ga}-q)=1}. \tag{6}$$

This is a finite sum of rational numbers, and yields Table 2.

$\gcd(h, 260)$	$\delta(h)$	$\gcd(h, 260)$	$\delta(h)$
1	541/1560	20	8/39
2	241/780	26	7/20
4	14/39	52	2/5
5	97/312	65	5/24
10	15/52	130	1/12
13	49/120	260	0

Table 2: Natural density of power floors coprime to 30, for every exponent.

All nonzero entries are at least 1/12, and only the row $g = 130$ attains that bound. This completes the density assertion of Theorem 2.1. For instance $h = 130$ fails the three-prime cover on exactly 1/12 of indices, whereas $h = 260$ has the cover on every index.

The certificate checks (6) by exact rational arithmetic. It independently computes the same densities using the matrix orbit of length dividing 780, and verifies agreement for every exponent residue class modulo 780. The analytic proof is what converts these finite orbit sums into natural densities; the certificate alone does not establish equidistribution.

7 Macroscopic repetitions in three arithmetic words

This section retains the coefficient sequences of the original constants. It adds the repetition statements of HTML T1–T3 to the prime-parity coding already present in the corpus [5]. The tools are classical: the prime number theorem, Siegel–Walfisz, and the divisor bound. No claim of a new transcendence principle or historical priority is made.

For a sequence $w = (w_n)_{n \geq 1}$, an *equal-block triple* is a triple of integers (r, s, L) with $0 \leq r < s$ and $L \geq 1$ such that

$$w_{r+i} = w_{s+i} \qquad (1 \leq i \leq L).$$

We say that w has *no macroscopic repetitions* if, for every $\varepsilon > 0$, there is $s_0(\varepsilon)$ such that no equal-block triple has $s \geq s_0(\varepsilon)$ and $L \geq \varepsilon s$. This definition concerns exact equality of the indicated coefficient blocks. It does not assert that the sequence has no short repetitions.

Let p_j be the jth prime, let $\pi(x)$ count primes at most x, and let $d(n)$ count the positive divisors of n.

Theorem 7.1 (Three repetition obstructions)**.** *Each of the following sequences has no macroscopic repetitions:*

1. $b_n^{(m)} = \pi(n) \bmod m$, *with residues in* $\{0, \dots, m-1\}$, *for every fixed integer* $m \geq 2$;

2. *the prime-gap sequence* $g_n = p_{n+1} - p_n$;

3. *the divisor-count sequence* $d(n)$.

We first isolate the exact analytic input in the first two assertions. The classical Siegel–Walfisz theorem states that, for every fixed $A, B > 0$, uniformly for $q \leq (\log X)^A$ and $(a, q) = 1$,

$$\pi(X; q, a) = \frac{\mathrm{Li}(X)}{\varphi(q)} + O_{A,B}\Big(\frac{X}{(\log X)^B}\Big). \tag{7}$$

Here $\pi(X; q, a)$ counts primes in the residue class a modulo q. The constants need not be effective for the arguments below. We also use $\vartheta(Y) = \sum_{p \leq Y} \log p \sim Y$ and $p_n \sim n \log n$. These are standard prime-number theorems; see [11].

Lemma 7.2 (A prime in the necessary reduced class)**.** *Fix* $0 < a < b$ *and* $C > 0$. *For every sufficiently large* X, *every prime* $\ell \leq C \log X$, *and every integer* D *with* $\ell \nmid D$, *there is a prime in* $[aX, bX]$ *congruent to* D *modulo* ℓ. *Moreover, if* $1 \leq D \leq 2X$, *then some prime* $\ell \ll \log X$ *does not divide* D; *the implied constant is absolute.*

Proof. Apply (7), with $A = 2$ and $B = 5$, at aX and bX. For large X its modulus range includes every $\ell \leq C \log X$. The difference of the main terms is

$$\frac{\mathrm{Li}(bX) - \mathrm{Li}(aX)}{\ell - 1} \gg_{a,b,C} \frac{X}{(\log X)^2},$$

whereas the total error is $O_{a,b}(X/(\log X)^5)$. The count is therefore positive. For the last assertion, if every prime at most $4 \log X$ divided D, then

$$D \geq \prod_{p \leq 4 \log X} p = \exp(\vartheta(4 \log X)) > X^2 > 2X$$

for large X, a contradiction. □

Proof of Theorem 7.1, part (1). Suppose that $b^{(m)}$ has equal blocks with $L \geq \varepsilon s$ and arbitrarily large s. Put $D = s - r$. Adjacent differences recover the prime indicator:

$$b_n^{(m)} - b_{n-1}^{(m)} \equiv \mathbf{1}_{\{n \text{ prime}\}} \pmod m.$$

Since an indicator is either 0 or 1 and $m \geq 2$, equality of adjacent residue differences implies equality of the indicators. Thus

$$\mathbf{1}_{\{r+i \text{ prime}\}} = \mathbf{1}_{\{s+i \text{ prime}\}} \qquad (2 \leq i \leq L). \tag{8}$$

Choose $c = \min(\varepsilon/4, 1/4) > 0$. The interval $[cs, 2cs]$ lies in $[2, L]$ for large s. As $1 \leq D \leq s$, choose by Lemma 7.2 a prime $\ell \ll \log s$ not dividing D. The same lemma gives a prime

$$y \in [(1 + c)s, (1 + 2c)s], \qquad y \equiv D \pmod \ell.$$

Write $i = y - s$. Then $2 \leq i \leq L$, while $r + i = y - D \geq cs > \ell$ and $r + i \equiv 0 \pmod \ell$. Consequently $r + i$ is composite and $s + i = y$ is prime, contradicting (8). □

Proof of Theorem 7.1, part (2). Suppose that $g_{r+i} = g_{s+i}$ for $1 \leq i \leq L$, where $L \geq \varepsilon s$ and s is arbitrarily large. Shortening both blocks if necessary, take $L = \lceil \varepsilon s \rceil$. Put $X = p_{s+1}$ and $D = p_{s+1} - p_{r+1}$, so $0 < D < X$. Summing the equal gaps gives

$$p_{r+j} + D = p_{s+j} \qquad (1 \leq j \leq L + 1). \tag{9}$$

The two lists contain all primes in their respective endpoint intervals. Translation by D therefore preserves primality for every integer in $[p_{r+1}, p_{r+L+1}]$.

The prime number theorem gives $p_{s+L+1}/X \longrightarrow 1+\varepsilon$. With $c = \min(\varepsilon/8, 1/8)$, the interval $[(1+c)X, (1+2c)X]$ lies strictly inside $[p_{s+1}, p_{s+L+1}]$ for large s. Choose a prime $\ell \ll \log X$ not dividing D, and then a prime y in that interior interval with $y \equiv D \pmod{\ell}$, using Lemma 7.2. Its translate satisfies

$$y - D \geq p_{r+1} + cX > \ell, \qquad y - D \equiv 0 \pmod{\ell}.$$

It is composite, contradicting the primality preservation just proved. □

Proof of Theorem 7.1, part (3). Suppose that divisor counts agree on two blocks as above, and shorten them to length $L = \lceil \varepsilon s \rceil$. Put $h = s - r$, so $1 \leq h \leq s$. The elementary identity

$$d(n) \equiv 1 \pmod{2} \quad \Longleftrightarrow \quad n \text{ is a square}$$

shows that every square k^2 in $[r+1, r+L]$ translates to a square $k^2 + h = j^2$. The number of squares in this interval is at least

$$\sqrt{r+L} - \sqrt{r} - 1 = \frac{L}{\sqrt{r+L} + \sqrt{r}} - 1 \gg_\varepsilon \sqrt{s}.$$

On the other hand, each such pair satisfies $(j-k)(j+k) = h$. There are at most $d(h)$ possibilities, since the positive factor $j-k$ determines $j+k$ and hence k. The classical divisor bound $d(n) = O_\eta(n^\eta)$ for every $\eta > 0$ [12], with $\eta = 1/4$, gives $d(h) = O(s^{1/4})$. This contradicts the displayed lower bound. □

Remark 7.3 (What the coefficient obstruction does and does not say). The natural generating series retain the original constants:

$$2\sum_{k\geq 1}(-1)^{k+1}2^{-p_k} = \sum_{n\geq 1} b_n^{(2)} 2^{-n}, \qquad \sum_{n\geq 1}\frac{1}{2^n - 1} = \sum_{n\geq 1} d(n)2^{-n}.$$

The first follows by summation of adjacent differences, and the second by absolutely convergent rearrangement of a geometric series. The divisor coefficients are not binary digits: carrying can change block structure. Theorem 7.1 excludes sufficient conditions requiring the specified long exact coefficient repetitions. It proves neither transcendence nor its impossibility by other methods. No assertion about a refined Diophantine exponent is needed here.

8 A classification of logarithmic contamination

For a rational function g satisfying $g(u) = O(u^{-3})$ at infinity, with poles only at negative integers, put

$$\mathcal{J}(g) = \int_0^\infty (\lfloor u \rfloor + 1) g(u)\, du.$$

The corpus already gives the complete pole-order evaluation of this transform [4]. HTML T9 adds a classification for simple poles and for two pole locations. We reproduce the needed transform calculation to make that classification independent of the longer source. Write $H_0 = 0$ and $H_n = \sum_{j=1}^n 1/j$.

Lemma 8.1 (Simple and double poles)**.** *Suppose*

$$g(u) = \sum_{p\geq 1}\left(\frac{\alpha_p}{u+p} + \frac{\beta_p}{(u+p)^2}\right), \qquad \alpha_p, \beta_p \in \mathbb{Q},$$

with finitely many nonzero coefficients and $g(u) = O(u^{-3})$. Then

$$\sum_p \alpha_p = 0, \qquad \sum_p \beta_p = \sum_p p\alpha_p, \tag{10}$$

$$\mathcal{J}(g) = \mathcal{L}(g) + \sum_p \beta_p(\gamma - H_{p-1}), \qquad \mathcal{L}(g) = \sum_p \alpha_p \log\Gamma(p). \tag{11}$$

Here γ is Euler's constant. In this specified decomposition the coefficient of γ is $B(g) = \sum_p \beta_p$.

Proof. The coefficients of u^{-1} and u^{-2} at infinity give (10). For a positive integer N, elementary summation over the intervals $[j, j+1]$ gives

$$A_p(N) := \int_0^N \frac{\lfloor u \rfloor + 1}{u+p}\, du = N\log(N+p) - \log\Gamma(N+p) + \log\Gamma(p),$$
$$B_p(N) := \int_0^N \frac{\lfloor u \rfloor + 1}{(u+p)^2}\, du = H_{N+p-1} - H_{p-1} - \frac{N}{N+p}.$$

Stirling's formula and $H_N = \log N + \gamma + o(1)$ yield

$$A_p(N) = N + (1/2 - p)\log N + p - \tfrac{1}{2}\log(2\pi) + \log\Gamma(p) + o(1),$$
$$B_p(N) = \log N + \gamma - H_{p-1} - 1 + o(1).$$

Multiply by the partial-fraction coefficients and add. The divergent terms cancel by (10), and $\sum p\alpha_p = \sum \beta_p$ cancels the remaining constant terms. Since the original weighted integral is absolutely convergent, taking $N \to \infty$ proves (11). □

Theorem 8.2 (Simple poles and the complete two-location case)**.** *The following statements hold.*

1. *Let $d \geq 3$, $1 \leq p_1 < \cdots < p_d$ be integers, and $C > 0$ be rational. For*

$$g(u) = \frac{C}{\prod_{i=1}^d (u + p_i)},$$

one has $B(g) = 0$ and

$$\mathcal{L}(g) > \frac{C}{(d-1)p_d^{d-1}} > 0.$$

2. *Suppose that the poles of g occur at exactly the two locations $-a, -b$, where $1 \leq a < b$ are integers, each with multiplicity at most two, and that $g(u) = O(u^{-3})$. Then*

$$\mathcal{L}(g) = \frac{\log\Gamma(b) - \log\Gamma(a)}{b-a}\, B(g). \tag{12}$$

Consequently, when $B(g) \neq 0$, its logarithmic part vanishes if and only if $(a, b) = (1, 2)$.

Proof. For (1), all β_p vanish and $\alpha_{p_i} = C/\prod_{j\neq i}(p_j - p_i)$. Let $f(x) = \log\Gamma(x)$ and write $[p_1, \ldots, p_d]f$ for its divided difference. The mean-value theorem for divided differences gives some $\xi \in (p_1, p_d)$ such that

$$\mathcal{L}(g) = C(-1)^{d-1}[p_1, \ldots, p_d]f = \frac{C(-1)^{d-1}}{(d-1)!} f^{(d-1)}(\xi).$$

For every integer $m \geq 2$ the convergent polygamma series is

$$f^{(m)}(x) = (-1)^m (m-1)! \sum_{n\geq 0} (n+x)^{-m}, \qquad x > 0.$$

Thus

$$\mathcal{L}(g) = \frac{C}{d-1}\sum_{n\geq 0}(n+\xi)^{-(d-1)} > \frac{C}{(d-1)p_d^{d-1}},$$

which proves the claim. These are classical identities; they also follow by differentiating the convergent product for Γ.

For (2), write the coefficients at the two locations as $\alpha_a, \alpha_b, \beta_a, \beta_b$. Equations (10) give

$$\alpha_b = -\alpha_a, \qquad B(g) = a\alpha_a + b\alpha_b = (b-a)\alpha_b.$$

Substitution into the definition of $\mathcal{L}(g)$ proves (12). At positive integer arguments, $\Gamma(1) = \Gamma(2) = 1$ and $\Gamma(n) = (n-1)!$ is strictly increasing for $n \geq 2$. The equality $\Gamma(a) = \Gamma(b)$ with $a < b$ therefore holds precisely for $(a, b) = (1, 2)$. □

Remark 8.3 (Scope of the classification). The words "coefficient" and "logarithmic part" refer to the explicit partial-fraction decomposition, not to a claimed linear independence of 1, γ, and logarithms. The exceptional pair gives, for example,

$$\mathcal{J}\Big(\frac{1}{(u+1)^2(u+2)}\Big) = \gamma, \qquad \mathcal{J}\Big(\frac{1}{(u+1)(u+2)^2}\Big) = 1-\gamma.$$

These are exact representations. They do not give a sequence of increasingly accurate rational approximations. Within the class of multiplicity at most two, a logarithm-free kernel with nonzero $B(g)$ and pole locations other than the fixed pair $\{1, 2\}$ needs at least three locations and at least one double pole. Higher pole orders and determinant constructions lie outside this classification.

9 The raw twin-prime Euler-product approximation

HTML T13 concerns the reduced denominator of the original finite Euler product. Its conclusion is an obstruction to that specific approximation sequence, independent of the twin-prime conjecture. Define

$$C_2 = \prod_{p>2}\left(1 - \frac{1}{(p-1)^2}\right) > 0, \qquad C_2(x) = \prod_{2<p\leq x}\left(1 - \frac{1}{(p-1)^2}\right).$$

Positivity and convergence follow from $\sum_{p>2}(p-1)^{-2} < \infty$ and positivity of every factor. For $x \geq 3$ write $C_2(x) = A_x/Q_x$ in lowest terms, with $Q_x > 0$.

Theorem 9.1 (Vanishing denominator-normalized exponent). *As $x \to \infty$,*

$$Q_x \geq 4^{\pi(x)-1}, \qquad C_2(x) - C_2 \sim \frac{C_2}{x\log x}, \tag{13}$$

$$\frac{-\log|C_2(x) - C_2|}{\log Q_x} \longrightarrow 0, \qquad Q_x^{\eta}|C_2(x) - C_2| \longrightarrow \infty \quad \textit{for every } \eta > 0. \tag{14}$$

Proof. Every numerator $p(p-2)$ in $C_2(x) = \prod_{2<p\leq x} p(p-2)/(p-1)^2$ is odd. Reduction of the product therefore removes none of the powers of two in its denominator. With v_2 the usual valuation,

$$v_2(Q_x) = 2\sum_{2<p\leq x} v_2(p-1) \geq 2(\pi(x)-1),$$

which proves the denominator bound.

For completeness, partial summation and the prime number theorem give

$$\sum_{p>x}\frac{1}{p^2}=-\frac{\pi(x)}{x^2}+2\int_x^\infty\frac{\pi(t)}{t^3}\,dt$$
$$\sim-\frac{1}{x\log x}+2\int_x^\infty\frac{dt}{t^2\log t}\sim\frac{1}{x\log x}.$$

Indeed the relative prime-number-theorem error is uniformly small for $t\ge x$ once x is large, and integration by parts gives $\int_x^\infty dt/(t^2\log t)\sim 1/(x\log x)$. Replacing p^{-2} by $(p-1)^{-2}$ changes the sum by $O(\sum_{n>x}n^{-3})=O(x^{-2})$. Therefore

$$R_x:=-\log\prod_{p>x}\left(1-\frac{1}{(p-1)^2}\right)=\sum_{p>x}\frac{1}{(p-1)^2}+O\left(\sum_{p>x}p^{-4}\right)\sim\frac{1}{x\log x}.$$

Since $C_2(x)=C_2e^{R_x}$, the error estimate in (13) follows.

Finally, $\log Q_x\ge(\pi(x)-1)\log 4\gg x/\log x$, whereas

$$-\log|C_2(x)-C_2|=\log x+\log\log x+O(1).$$

Their ratio tends to zero. For fixed $\eta>0$, the same bounds give

$$\log(Q_x^\eta|C_2(x)-C_2|)\ge\eta(\pi(x)-1)\log 4-\log x-\log\log x+O(1)\longrightarrow\infty,$$

proving the final assertion. □

Remark 9.2 (The route excluded). If $C_2=u/v$ were rational, rational separation would give $|C_2(x)-C_2|\ge 1/(vQ_x)$; the difference is nonzero since the omitted Euler factors are strictly below one. Theorem 9.1 shows that the actual error is much larger than this lower bound. The same observation applies to any fixed power of the denominator. It does not decide the arithmetic nature of C_2 and does not exclude weighted products or combinations of several truncations.

10 Cross-index elimination for an asymmetric Euler integral

This section preserves a useful synthesis from the HTML research record. The exact integral family and its logarithmic obstruction are already in manuscript [001] [4], while the moving-window counting mechanism is already in manuscript [021] [7]. Combining them gives the following precise statement, with the nonvanishing issue left explicit. It is not presented as a new proof mechanism or as an irrationality result for Euler's constant.

For $n\ge 1$, define

$$J_n=\int_0^1\int_0^1\frac{[x(1-x)y(1-y)]^n(1-x)}{(1-xy)(-\log(xy))}\,dx\,dy,\qquad d_n=\operatorname{lcm}(1,\dots,2n+2).\tag{15}$$

Write $H_k=\sum_{j=1}^k 1/j$, $H_0=0$, and let γ denote Euler's constant. All logarithms below are natural.

Theorem 10.1 (Exact form and cross-index elimination)**.** *There are integers $a_n>0$, b_n, and $c_{n,\ell}\ge 0$, indexed by primes $\ell\le 2n+1$, such that*

$$d_nJ_n=a_n\gamma+b_n+\sum_{\substack{\ell\le 2n+1\\ \ell\text{ prime}}}c_{n,\ell}\log\ell,\qquad a_n=d_n\binom{2n+1}{n}.\tag{16}$$

The logarithmic sum is strictly positive. With $c = \log 16 - 2 > 0$,

$$0 < d_n J_n \le \exp(-cn + o(n)), \tag{17}$$

$$\log \max(1, a_n, |b_n|, \max_{\ell} c_{n,\ell}) \le (2 + \log 4)n + o(n). \tag{18}$$

For every sufficiently large N, *there is a nonzero integer vector* $(u_N, \dots, u_{2N})$ *with*

$$\log \max(1, \max_{N \le n \le 2N} |u_n|) = O(N/\log N) \tag{19}$$

which eliminates every logarithm in the moving window:

$$\sum_{n=N}^{2N} u_n d_n J_n = Q_N \gamma + P_N, \quad P_N, Q_N \in \mathbb{Z}, \quad |Q_N \gamma + P_N| \le \exp(-cN + o(N)). \tag{20}$$

Every nonzero eliminating vector has both positive and negative entries. Moreover, there also exist nonzero vectors satisfying (19) *for which* $P_N = Q_N = 0$ *exactly. Consequently, the assertion that the vector is nonzero cannot be substituted for the assertion that its resulting linear form is nonzero.*

10.1 Exact coefficients from a half-line kernel

The identities

$$\frac{1}{-\log(xy)} = \int_0^\infty (xy)^t \, dt, \qquad \frac{1}{1-xy} = \sum_{j \ge 0} (xy)^j$$

have nonnegative integrands. Tonelli's theorem, followed by the change of variable $u = t + j$, therefore gives

$$J_n = \int_0^\infty (\lfloor u \rfloor + 1) g_n(u) \, du, \tag{21}$$

$$g_n(u) = B(u+n+1, n+2) B(u+n+1, n+1) = \frac{n!(n+1)!}{(u+M) \prod_{p=L}^{M-1} (u+p)^2}, \tag{22}$$

where $L = n+1$, $M = 2n+2$, and $B(s,t) = \int_0^1 x^{s-1}(1-x)^{t-1} \, dx$. The factor $\lfloor u \rfloor + 1$ counts the nonnegative integers $j \le u$; its values at integer endpoints do not affect the integral.

The partial-fraction expansion is

$$g_n(u) = \sum_{p=L}^{M} \frac{\alpha_p}{u+p} + \sum_{p=L}^{M-1} \frac{\beta_p}{(u+p)^2}, \tag{23}$$

where

$$\beta_{L+k} = \binom{n}{k} \binom{n+1}{k}, \qquad 0 \le k \le n, \tag{24}$$

$$\alpha_{L+k} = \beta_{L+k} \left(2H_k - 2H_{n-k} - \frac{1}{n+1-k} \right), \qquad \alpha_M = \frac{1}{n+1}. \tag{25}$$

Indeed, evaluating $(u+L+k)^2 g_n(u)$ at $u = -L-k$ gives

$$\beta_{L+k} = \frac{n!(n+1)!}{(n+1-k)(k!)^2((n-k)!)^2},$$

and logarithmic differentiation gives the corresponding α_{L+k}. The remaining simple residue gives α_M. Comparing the first two terms at infinity in (23) gives

$$\sum_p \alpha_p = 0, \qquad \sum_p p\alpha_p = \sum_p \beta_p. \tag{26}$$

Applying the common-cutoff transform in Lemma 8.1 to (23) gives

$$J_n = \binom{2n+1}{n}\gamma - \sum_{p=L}^{M-1} \beta_p H_{p-1} + \sum_{p=L}^{M} \alpha_p \log((p-1)!). \tag{27}$$

Here Vandermonde's identity gives $\sum_{k=0}^{n} \binom{n}{k}\binom{n+1}{k} = \binom{2n+1}{n}$. In particular, the coefficient of γ is exact, not asymptotic.

Let $v_\ell(m)$ be the exponent of a prime ℓ in the positive integer m. Explicit coefficients for (16) are

$$b_n = -d_n \sum_{p=L}^{M-1} \beta_p H_{p-1}, \tag{28}$$

$$c_{n,\ell} = d_n \sum_{p=L}^{M} \alpha_p v_\ell((p-1)!) = d_n \sum_{j=L}^{M-1} T_j v_\ell(j), \qquad T_j = \sum_{p=j+1}^{M} \alpha_p. \tag{29}$$

The last equality follows by expanding the factorial logarithms and using $\sum_p \alpha_p = 0$ for the terms $j < L$. Every denominator in α_p divides $\operatorname{lcm}(1, \ldots, n+1)$, and every denominator in H_{p-1} divides d_n. Since the β_p are integers, (28)–(29) are integers. Their prime support is contained in $\ell \le 2n+1$, without any assumption about the kernel's rank. For example,

$$12J_1 = 36\gamma - 48 + 30\log 2 + 6\log 3, \tag{30}$$

$$60J_2 = 600\gamma - 1125 + 760\log 2 + 200\log 3 + 20\log 5. \tag{31}$$

To prove positivity, set $e_k = 2H_k - 2H_{n-k} - 1/(n+1-k)$. Then $e_0 < 0$, $e_n > 0$, and

$$e_{k+1} - e_k = \frac{2}{k+1} + \frac{1}{n-k} + \frac{1}{n+1-k} > 0 \qquad (0 \le k < n).$$

Since $\beta_{L+k} > 0$ and $\alpha_M > 0$, the simple residues have one change from negative to positive, allowing a zero at the transition. Their total is zero. Every proper tail T_j in (29) is therefore strictly positive. It follows that $c_{n,\ell} \ge 0$ and

$$\sum_{\ell} c_{n,\ell} \log \ell = d_n \sum_{j=L}^{M-1} T_j \log j > 0. \tag{32}$$

10.2 Decay, arithmetic size, and the counting argument

The elementary inequalities $-\log(xy) \ge 1 - xy \ge 1 - x$ imply

$$0 < J_n \le \int_0^1 \int_0^1 x^n (1-x)^{n-1} y^n (1-y)^n \, dx\, dy \le 2\,16^{-n}.$$

For the last bound, use $[x(1-x)]^{n-1} \le 4^{-(n-1)}$ and $[y(1-y)]^n \le 4^{-n}$, and integrate the remaining factor x. The standard prime number theorem in the form

$$\log \operatorname{lcm}(1, \ldots, m) = m + o(m)$$

gives $\log d_n = 2n + o(n)$, proving (17). Also $\beta_p \le 2 \cdot 4^n$, $|\alpha_p| \le 2 \cdot 4^n(4H_n + 1)$, and $v_\ell((p-1)!) = O(n \log(n+1))$ uniformly in p, ℓ. Equations (28)–(29), with at most $n+2$ terms, prove (18). Polynomial factors contribute only $O(\log(n+1))$ to its logarithm.

We record the precise elementary counting step. If an integer matrix D has $m < k$ rows and columns respectively and entries of absolute value at most $B \ge 1$, put

$$H = \left\lfloor (3kB)^{m/(k-m)} \right\rfloor + 1. \tag{33}$$

The $(H+1)^k$ vectors in $\{0,\dots,H\}^k$ have at most $(2kBH+1)^m \le (3kBH)^m$ different images under D. Because $H^{k-m} > (3kB)^m$, two of those vectors have the same image. Their difference is a nonzero integer vector w with $Dw = 0$ and $\max_i |w_i| \le H$. This proof requires no rank hypothesis and no nonvanishing hypothesis on any associated linear form.

Apply this to the matrix

$$D_N = (c_{n,\ell})_{\substack{\ell \le 4N+1 \text{ prime},\\ N \le n \le 2N}},$$

where coefficients outside their individual prime supports are zero; primes index rows and n indexes columns. It has $k = N+1$ columns and $m = \pi(4N+1) = O(N/\log N)$ rows. The preceding height estimate gives $\log B = O(N)$, so (33) gives

$$\log H \le \log 2 + \frac{m}{k-m}\log(3kB) = O(N/\log N).$$

Let u be the resulting kernel vector. Summing (16) cancels the logarithms, with $Q_N = \sum u_n a_n$ and $P_N = \sum u_n b_n$. Uniformly for $N \le n \le 2N$, $d_n J_n \le \exp(-cN + o(N))$; consequently

$$\left| \sum_{n=N}^{2N} u_n d_n J_n \right| \le (N+1) H \max_{N \le n \le 2N} d_n J_n \le \exp(-cN + o(N)).$$

This proves the existence and size statements in the theorem.

Finally, take the row vector with entries $\log \ell$. By (32), every coordinate of its product with D_N is strictly positive. Hence $D_N u = 0$ is impossible for a nonzero vector whose coordinates all have the same weak sign. The positivity of J_n therefore supplies no positivity of the compressed form. To make the distinction still more explicit, append the two integer rows (a_n) and (b_n) to D_N. Their entries also have size $\exp(O(N))$, and the row count is still $O(N/\log N)$. The same counting argument produces a nonzero integer vector v of the same height with

$$D_N v = 0, \qquad \sum_{n=N}^{2N} v_n a_n = 0, \qquad \sum_{n=N}^{2N} v_n b_n = 0.$$

Thus $\sum v_n d_n J_n = 0$ exactly. This completes the proof of Theorem 10.1 and identifies the missing step in any proposed irrationality application: one must produce infinitely many compressed forms that are provably nonzero. No such claim is made here.

11 Exact coordinates and finite-system obstructions for two reciprocal sums

The Erdős–Borwein constant and the reciprocal Fibonacci sum are

$$E = \sum_{n \ge 1} \frac{1}{2^n - 1}, \qquad \Psi = \sum_{n \ge 1} \frac{1}{F_n}, \qquad F_1 = F_2 = 1, \quad F_{n+2} = F_{n+1} + F_n.$$

We retain these complete sums. They are values of the same elementary meromorphic family. The family has a first-order multiplicative shift equation, but cannot be a component of a finite rational power-Mahler system. We prove that distinction and the associated orbit identities explicitly.

This section consolidates statements T6, T27 and T28 of the supplied HTML. Their scalar and vector obstructions are one result and its linear-algebra consequence, rather than independent discoveries. The supplied corpus already contains the integer-base shift identity in manuscript [030] [8], equation `p04:ci20:shift`; the jet formula below follows by differentiating and iterating that type of identity. The arguments are elementary specializations of established methods. No historical priority or transcendence of either displayed constant is asserted.

11.1 One family, with the original constants preserved

For a complex number Q with $|Q| > 1$, define

$$L_Q(z) = \sum_{n\geq 1} \frac{z^n}{Q^n - 1}, \qquad |z| < |Q|. \tag{34}$$

Absolute convergence follows by comparison with $\sum_{n\geq 1}(|z|/|Q|)^n$. Put $\phi = (1+\sqrt{5})/2$.

Lemma 11.1 (Exact value coordinates)**.** *The identities*

$$E = L_2(1), \qquad \Psi = \sqrt{5}\, L_{-\phi^2}(-\phi) \tag{35}$$

hold with absolute convergence. The functions satisfy the meromorphic identity

$$L_Q(Qz) - L_Q(z) = \frac{z}{1-z}. \tag{36}$$

Moreover,

$$L_Q(z) = \sum_{m\geq 1} \frac{z}{Q^m - z} \tag{37}$$

gives a meromorphic continuation to $\mathbb{C}$. Its poles are exactly $Q, Q^2, Q^3, \ldots$, each simple, with residue $-Q^m$ at Q^m.

Proof. The first equality in (35) is the definition. Binet's formula gives

$$\sqrt{5}F_n = \phi^n - (-1)^n\phi^{-n}, \qquad \frac{(-\phi)^n}{(-\phi^2)^n - 1} = \frac{\phi^n}{\phi^{2n} - (-1)^n} = \frac{1}{\sqrt{5}F_n}.$$

Since $\phi < \phi^2$, both series representations converge absolutely. For $|z| < 1$, subtraction of the series proves (36). Also, for $|z| < |Q|$,

$$\frac{1}{Q^n - 1} = \sum_{m\geq 1} Q^{-mn}.$$

The absolute double sum is finite, so summing first in n gives (37). On any compact set, the tail of the latter sum is bounded by a constant times $\sum_m |Q|^{-m}$. It is therefore normally convergent away from the listed poles. Their absolute values are distinct, so only one summand is singular at each pole; its residue is $-Q^m$. Analytic continuation now proves the shift identity on its entire meromorphic domain. □

11.2 Logarithmic derivatives and the exact orbit cocycle

The product

$$\mathcal{E}_Q(x) = \prod_{m\geq 1} (1 + x/Q^m) \tag{38}$$

converges normally on compact sets and defines an entire function. Its zeros are the simple points $-Q^m$, $m \geq 1$. Its logarithmic derivative is

$$R_Q(x) = \frac{\mathcal{E}_Q'(x)}{\mathcal{E}_Q(x)} = \sum_{m\geq 1} \frac{1}{Q^m + x}. \tag{39}$$

These assertions follow from locally uniform convergence of $\sum_m |x/Q^m|$ and of the differentiated logarithms away from their zeros. In particular,

$$L_Q(z) = zR_Q(-z), \qquad E = R_2(-1), \qquad \Psi = -\sqrt{5}\phi R_{-\phi^2}(\phi). \tag{40}$$

For an integer $s \geq 0$ write $R_Q^{(s)}$ for the ordinary sth derivative. Set $c_s = (-1)^s s!$, and for every integer k define the rational function

$$C_{k,s}(x) = \begin{cases} \displaystyle\sum_{j=0}^{k-1} \frac{Q^{j(s+1)}}{(1+Q^j x)^{s+1}}, & k > 0, \\ 0, & k = 0, \\ \displaystyle -\sum_{j=k}^{-1} \frac{Q^{j(s+1)}}{(1+Q^j x)^{s+1}}, & k < 0. \end{cases} \tag{41}$$

Theorem 11.2 (All derivative orders and both orbit directions)**.** *For $|Q| > 1$, $s \geq 0$ and $k \in \mathbb{Z}$, the meromorphic identity*

$$Q^{k(s+1)} R_Q^{(s)}(Q^k x) = R_Q^{(s)}(x) + c_s C_{k,s}(x) \tag{42}$$

holds. Suppose additionally that $Q, x \in \overline{\mathbb{Q}}$ and that the values under consideration are finite. The regular values of a fixed derivative order on this orbit belong to

$$\operatorname{span}_{\overline{\mathbb{Q}}}\{1, R_Q^{(s)}(x)\}.$$

Consequently all regular orbit values of orders $0 \leq s \leq S$ span a space of dimension at most $S+2$ over $\overline{\mathbb{Q}}$.

Proof. Separating the first factor of the product gives

$$\mathcal{E}_Q(Qx) = (1+x)\mathcal{E}_Q(x), \qquad QR_Q(Qx) = R_Q(x) + \frac{1}{1+x}.$$

Differentiating s times yields

$$Q^{s+1} R_Q^{(s)}(Qx) = R_Q^{(s)}(x) + \frac{c_s}{(1+x)^{s+1}}.$$

For $j \in \mathbb{Z}$, substitute $Q^j x$ and multiply by $Q^{j(s+1)}$. The left side minus its first right-side term is the difference of consecutive terms of the sequence $Q^{j(s+1)} R_Q^{(s)}(Q^j x)$. Telescoping forward or backward proves (42).

For finite evaluation at algebraic Q, x, each term of $C_{k,s}(x)$ is algebraic provided its denominator is nonzero. This is automatic when both endpoint values in (42) are regular. Indeed a vanishing denominator means $x = -Q^{-j}$ for an intermediate index j; in either direction one endpoint then has the form $-Q^m$ with $m \geq 1$ and is a pole. Thus every regular value is an algebraic multiple of the base value plus an algebraic number. Taking the indicated spans proves the two bounds. □

Remark 11.3 (Affine information and the pole qualification)*.* Modulo the additive subspace $\overline{\mathbb{Q}}$, the normalized values $Q^{k(s+1)} R_Q^{(s)}(Q^k x)$ all have the same class. This is the precise sense in which a fixed derivative order supplies at most one new orbit coordinate. It does not assert that the coordinate is transcendental, or that all determinants assembled in arbitrary ways from orbit data vanish.

The qualification about poles matters for the original targets. For $E = R_2(-1)$ every forward point -2^k, $k \geq 1$, is a pole. Its regular backward orbit is -2^{-k}, $k \geq 0$; for example,

$$R_2(-1/2) = 2R_2(-1) - 2 = 2E - 2.$$

For $Q = -\phi^2$ and $x = \phi$, every integer orbit point is regular: an equality $Q^k \phi = -Q^m$ would imply $\phi^{2k+1} = \phi^{2m}$, impossible for integers k, m.

11.3 A complete obstruction to finite power-Mahler embedding

A power-Mahler equation uses the substitution $z \mapsto z^d$ for an integer $d \geq 2$. It is distinct from the multiplicative shift $z \mapsto Qz$ in (36). The pole sets make that distinction exact here.

Theorem 11.4 (Independence of all power iterates)**.** *Let $Q \in \mathbb{C}$ satisfy $|Q| > 1$, and let $d \geq 2$ be an integer. For every $r \geq 0$, the functions*

$$1,\ L_Q(z),\ L_Q(z^d), \ldots, L_Q(z^{d^r})$$

are linearly independent over $\mathbb{C}(z)$. Equivalently, if $a_0, \ldots, a_r, b \in \mathbb{C}(z)$ and

$$\sum_{j=0}^{r} a_j(z) L_Q(z^{d^j}) = b(z) \tag{43}$$

holds as an analytic or meromorphic germ, then every coefficient and b vanish identically.

Proof. Assume some a_j is nonzero, and let r now denote its greatest index. For every positive integer m with $d \nmid m$, choose any z_m satisfying

$$z_m^{d^r} = Q^m.$$

The numbers z_m are nonzero and have distinct absolute values $|Q|^{m/d^r}$. By Lemma 11.1, the highest iterate $L_Q(z^{d^r})$ has a simple pole at each z_m; the derivative of z^{d^r} there is nonzero.

No lower iterate has a pole there. Otherwise for some $j < r$ and $n \geq 1$ one would have $z_m^{d^j} = Q^n$. Raising to d^{r-j} gives

$$Q^m = Q^{nd^{r-j}}.$$

Taking absolute values, using $|Q| > 1$, implies $m = nd^{r-j}$ and hence $d \mid m$, a contradiction. This argument also covers $r = 0$, for which there are no lower iterates.

Exclude the finitely many zeros and poles of the nonzero rational function a_r and the finitely many poles of $b, a_0, \ldots, a_{r-1}$. Infinitely many z_m remain. At any one of them the left side of (43) has a pole from exactly its highest term, whereas the right side is regular. The meromorphic continuation of the germ identity is therefore impossible. All a_j vanish, and then so does b. □

Remark 11.5 (Negative and complex bases are included). The proof uses absolute values to compare powers of Q and never requires $Q > 0$. In particular it applies to $Q = -\phi^2$ without a choice of logarithm. An alternative explicit construction for real Q and $r \geq 1$ is to choose a primitive d^rth root ζ and set $z_t = \zeta |Q|^{2t}$. The highest iterate has its pole at Q^{2td^r}. A pole in a lower iterate would require the even index $n = 2td^j$, hence a positive pole Q^n, while $\zeta^{d^j} \neq 1$. This gives the same contradiction.

Corollary 11.6 (No finite rational affine system)**.** *There is no finite vector Y of analytic or meromorphic germs, having L_Q as a component, that satisfies*

$$Y(z^d) = A(z)Y(z) + B(z), \qquad A \in \mathrm{Mat}_n(\mathbb{C}(z)), \quad B \in \mathbb{C}(z)^n.$$

The assertion holds even without assuming that A is invertible.

Proof. Adjoin the constant component 1. Iteration of the displayed system expresses every component of $Y(z^{d^j})$ in the fixed $\mathbb{C}(z)$-vector space generated by $1, Y_1(z), \ldots, Y_n(z)$. It follows that all the iterates of the component L_Q lie in a space of dimension at most $n+1$. Sufficiently many of them are linearly dependent, contrary to Theorem 11.4. □

11.4 What these statements leave unresolved

The value identities (35) and (40) preserve the two named constants exactly. The orbit calculation controls information at points related by one fixed multiplicative shift. The last theorem excludes a finite rational power-Mahler system containing the specified function. It does not exclude a different function that happens to take the same value at one point, and it does not exclude other arithmetic methods.

No value-transcendence inference follows merely from functional independence. For example, the same family has the algebraic value $L_Q(0) = 0$, and for algebraic Q its logarithmic derivative has

$$R_Q(0) = \sum_{m \geq 1} Q^{-m} = \frac{1}{Q-1} \in \overline{\mathbb{Q}}.$$

Proving that $R_2(-1)$ or $R_{-\phi^2}(\phi)$ is transcendental requires a theorem for those particular values that is not supplied by the pole argument or the jet cocycle. The unconditional output of this section is a precise restriction on two proposed functional routes, with full proofs and unchanged target values.

12 Consequences and remaining research tasks

The conclusions have different logical strengths. For the quartic Salem number and the specified prime set, the condition $260 \mid h$ is necessary and sufficient, and the density table is complete. The repeated-block and pole arguments are also unconditional theorems with the exact domains stated in their hypotheses.

For Euler's constant, cross-index elimination removes every logarithm with controlled integer coefficients. However, appending only two more rows to the same integer matrix already produces comparably short vectors whose resulting form is exactly zero. An irrationality argument therefore needs a mechanism that proves nonvanishing for infinitely many selected forms; vector nontriviality and positivity of the individual integrals cannot provide it.

For the two reciprocal sums, the meromorphic family and the original values are unchanged. Its multiplicative-shift jets contain at most one new coordinate per derivative order, and its power iterates cannot lie in any finite rational affine system. These conclusions restrict specified approaches and supply no assertion about every possible representation of the same value.

Finally, the twin-prime partial-product theorem concerns the exact unweighted rational truncations in its statement. Weighted products, other sequences of approximants, and unrelated methods are outside that theorem. Keeping each conclusion at its proved scope is necessary for this integrated manuscript to be reusable as a research starting point.

A A self-contained exact arithmetic certificate

The listing below is part of this manuscript. It can be copied unchanged to a text file and run with Python 3; it needs only the standard library. It checks the trace period, the unsafe-divisor witnesses, every exponent class modulo the trace period, the complete rational density table, and an independent quotient-ring computation of the matrix order and every density for all 780 exponent classes. Thus no separate code file is required to reproduce the finite part of the proof. The strict correction window and the equidistribution argument in the main text supply the infinite-index implications; those analytic results are not inferred from this finite calculation.

The quotient-ring computation uses

$$X^4 = X^3 + X^2 + X - 1 \quad \text{in } (\mathbb{Z}/30\mathbb{Z})[X]/(P(X)).$$

Multiplication by X in the basis $1, X, X^2, X^3$ is exactly the companion matrix displayed in the main text. Consequently its order is the order computed below. For a representative $r_0 + r_1X + r_2X^2 + r_3X^3$, the trace of its multiplication matrix is $4r_0 + r_1 + 3r_2 + 7r_3$ modulo 30; this follows by taking the traces of I, C, C^2, C^3. The recurrence state transition is invertible, so the first repeated initial four-term state proves the exact trace period for all indices.

Listing 1: Exact certificate for the quartic classification.

```
from fractions import Fraction as F
from math import gcd

M = 30
T = [4, 1, 3, 7]
for n in range(4, 264):
    T.append((T[-1] + T[-2] + T[-3] - T[-4]) % M)
period = next(k for k in range(1, 261)
              if T[k:k+4] == T[:4])
assert period == 260

Q = (-1, 0, 1, 2)
safe = {s for s in range(M)
        if all(gcd(s-q, M) > 1 for q in Q)}
assert safe == {4, 5, 26, 27}
divisors = [g for g in range(1, 261) if 260 % g == 0]
expected = {
    1: F(541,1560), 2: F(241,780), 4: F(14,39),
    5: F(97,312), 10: F(15,52), 13: F(49,120),
    20: F(8,39), 26: F(7,20), 52: F(2,5),
    65: F(5,24), 130: F(1,12), 260: F(0)
}
assert divisors == list(expected)
assert [T[g] for g in divisors[:-1]] == [
    1, 3, 7, 16, 18, 1, 22, 9, 19, 16, 24
]
weights = {-1: F(1,3), 0: F(1,6),
           1: F(1,6), 2: F(1,3)}
for g in divisors:
    orbit = range(0, period, g)
    value = sum((weights[q] for r in orbit for q in Q
                 if gcd(T[r]-q, M) == 1), F(0))
    value /= len(orbit)
    assert value == expected[g]
    assert (all(T[r] in safe for r in orbit)) == (g == 260)
    print(g, value)

for h in range(1, 261):
    residues = {h*n % period for n in range(period)}
    covered = all(T[r] in safe for r in residues)
```

```
    assert covered == (h % 260 == 0)
assert min(v for v in expected.values() if v) == F(1,12)
assert [g for g, v in expected.items() if v == F(1,12)] == [130]

# Independent multiplication in (Z/30Z)[X]/(P).
def mul(a, b):
    c = [0] * 7
    for i in range(4):
        for j in range(4):
            c[i+j] += a[i] * b[j]
    for k in range(6, 3, -1):
        v = c[k]
        c[k-1] += v
        c[k-2] += v
        c[k-3] += v
        c[k-4] -= v
    return tuple(v % M for v in c[:4])

one, x = (1,0,0,0), (0,1,0,0)
r = one
orbit = [one]
for n in range(1, 781):
    r = mul(r, x)
    if n == 260:
        assert r == (21,20,20,20)
    if n == 520:
        assert r == (11,10,10,10)
    assert (r == one) == (n == 780)
    if n < 780:
        orbit.append(r)
assert len(set(orbit)) == 780

# Matrix traces from the quotient orbit, not from T.
basis_traces = (4, 1, 3, 7)
matrix_traces = [sum(a*b for a, b in zip(r, basis_traces))
                 % M for r in orbit]
for h in range(1, 781):
    length = 780 // gcd(h, 780)
    traces = [matrix_traces[h*k % 780]
              for k in range(length)]
    value = sum((weights[q] for s in traces for q in Q
                 if gcd(s-q, M) == 1), F(0)) / length
    assert value == expected[gcd(h, 260)]
    assert all(s in safe for s in traces) == (h % 260 == 0)
print("PASS: trace period 260, matrix order 780,")
print("all 780 matrix-orbit density classes verified.")
```

All numerical quantities in the listing are integers or reduced rational numbers. No numerical approximation to τ, its conjugates, or their angles is used. In particular, the listing does not test primality and does not identify the terms coprime to 30 with prime terms.

B Source concordance for the selected results

The labels T1–T60 refer to the supplied HTML record, not to new theorem numbers in this paper. Corpus identifiers refer to the numbered files at the immutable snapshot in Section 1. Additional below means that the same substantive statement was not located in that snapshot; it is not a worldwide priority assertion.

Source	Retained mathematical content	Relation to the corpus
T1–T3	No macroscopic repeated blocks in the prime-count residue word, prime-gap word and divisor-count word.	Additional block statements. Manuscript [003] contains the prime codings, while [030] studies related Lambert functions. We state block theorems directly, avoiding an unverified transfer to the binary digits after carrying.
T9	Simple-pole logarithmic positivity and the two-location classification.	Extends the complete pole transform in [001], `prop:complete-pole-transducer`. The transform itself is reproduced and credited.
T13	Reduced denominator versus tail for the raw twin-prime Euler product.	No same substantive result located. The proof uses the elementary two-adic denominator bound and the prime number theorem.
T55	Short cross-index logarithm elimination for the asymmetric Euler integral, including the zero-form obstruction.	Synthesis: the exact integral and its arithmetic occur in [001], `thm:Jn-asymptotic-integerization`; the moving-window counting mechanism already occurs in [021] for a different integral family.
T6, T27, T28	Fixed-Q Lambert poles, the all-jet orbit identity, and no finite rational power-Mahler embedding.	The integer-base shift equation already occurs in [030], `p04:ci20:shift`. The fixed-Q pole proof differs from its variable-base Lambert theorem. Scalar and vector obstructions are combined into one result and its corollary.
T60 and the preceding Salem note	The strict correction mechanism, prescribed-exponent criterion, quartic equivalence $260 \mid h$, and all twelve exact densities.	The norm window, independent angles, existence construction and sharp prime-cover sets occur in [005]/[031]. The preceding note adds the fixed-exponent classification and its density table; these are retained in full.

The comparison was performed across all 49 TeX files, rather than by matching titles alone. Nineteen of T41–T60 have substantive matches in the corpus; T55 is a literal-family synthesis of existing methods. Other omitted items include already incorporated chamber-moment and denominator results in [011], and the conditional Mills identities in [010]/[031]. Their omission is an editorial decision to avoid reprinting existing material as an additional result.

Two scope repairs are reflected in the mathematical text. First, the forward orbit of $R_2(-1)$ consists of poles, so finite-value orbit identities for the Erdős–Borwein constant use its backward orbit. Second, the quartic trace period is 260, whereas the companion matrix order modulo 30 is 780. The embedded certificate checks both quantities separately and confirms the same densities by both representations.

References

[1] Supplied research record, *Transcendence full-proof research audit* (HTML, Korean), statements T1–T60, supplied uploaded revision. Primary working material supplied for this integration; the labels in Appendix B identify the retained statements.

[2] *Fixed-Exponent Prime Covers for Salem Power Floors: An exact quartic exponent and density classification*, research note prepared for KEIO, 24 October 2026. The complete mathematical argument is retained in this manuscript.

[3] QEEDDD, *Verifiable Mathematical Reasoning Corpus*, 49 numbered TeX research manuscripts, immutable revision `f70e5fedabf7b90313f703eb85f7dff72a055541`, checked 24 October 2026 (Korea Standard Time). Repository snapshot. The following numbered items refer to this same revision.

[4] QEEDDD, corpus [001], *Exponential-Integral Periods across Arithmetic, Pade, Hankel, Reciprocal Geometry, and Frobenius Framing*, especially the complete pole transform and the asymmetric Euler-integral family.

[5] QEEDDD, corpus [003], *Integrated Structures and Boundary Spectra in Sparse Prime Codings*.

[6] QEEDDD, corpus [005], *Composite Integer Parts of Powers of Algebraic Units (Pure Cubic Fields, Salem Numbers, and a K3 Application)*.

[7] QEEDDD, corpus [021], *Arithmetic Alternatives for Euler's Constant*. Moving-window integer-kernel elimination of logarithmic coefficients.

[8] QEEDDD, corpus [030], *Arithmetic and Observation: Classifications, Recovery, and Transcendence Barriers. Unified Edition of Twelve Successor Manuscripts*, especially equation `p04:ci20:shift`.

[9] QEEDDD, corpus [031], *Prime-Representing Constants and Related Structures (Encodings, Arithmetic, Geometry, and Modular Obstructions)*, section on sharp finite prime covers for algebraic power floors.

[10] S. Hahn, D. Ismailescu, G. Kim and M. Kim, *Explicit Algebraic Numbers all whose Integer Parts of Powers are Composite*, arXiv:2608.05309v1 (2026), especially Section 7. `https://arxiv.org/abs/2608.05309`.

[11] H. Davenport, *Multiplicative Number Theory*, third edition, revised by H. L. Montgomery, Graduate Texts in Mathematics 74, Springer, 2000.

[12] G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, sixth edition, revised by D. R. Heath-Brown and J. H. Silverman, Oxford University Press, 2008.